CANDIDATE NAME			
CLASS	2T	INDEX NUMBER	

BIOLOGY

8875/01

Paper 1 Multiple Choice

Friday 18 September 2009
1 hour

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

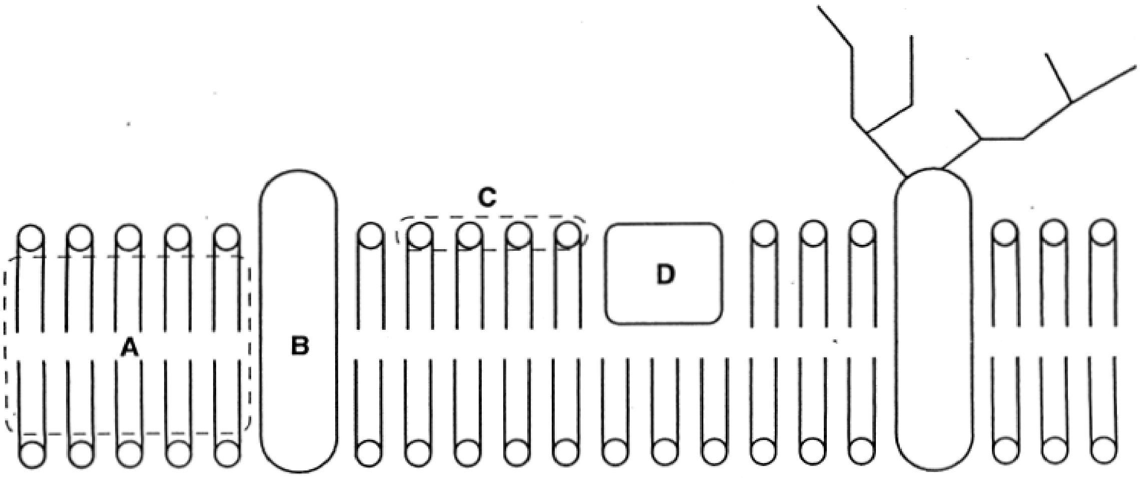
Any rough working should be done in this booklet.

This document consists of **12** printed pages.

[Turn over]

1 The diagram shows a model of the structure of a biological membrane.

Which part would restrict the movement of small, lipid-insoluble molecules?



Nov 02 Q1 Ans A

2 A certain cell surface membrane is made entirely of phospholipid and is 5 nm thick. A volume of 1 mm^3 of this phospholipid membrane was homogenised and dropped onto the surface of water in a large tray. The phospholipid spread out to form a continuous thin film.

What is the expected area of this film?

A 200 000 mm^2 because the molecules form a single layer.
B 200 000 mm^2 because the molecules form a double layer.
C 400 000 mm^2 because the molecules form a single layer.
D 400 000 mm^2 because the molecules form a double layer.

Nov 05 Q4 Ans C

3 What correctly describes the structure of starch?

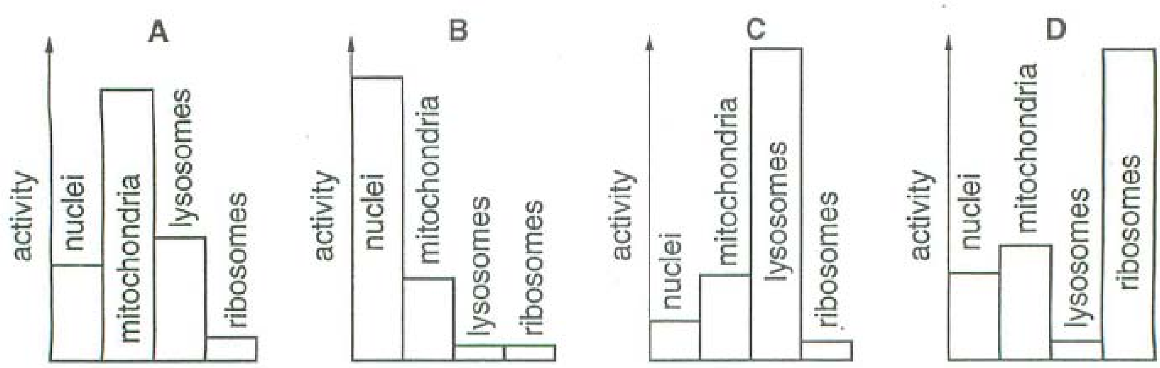
	α glucose	β glucose	1,4 glycosidic bonds	1,6 glycosidic bonds	molecules form coiled chains
A	✓	✗	✓	✓	✓
B	✗	✓	✓	✗	✗
C	✗	✓	✗	✓	✓
D	✓	✗	✓	✓	✗

key
✓ present
✗ absent

Nov 08 (H1) Q20 Ans A

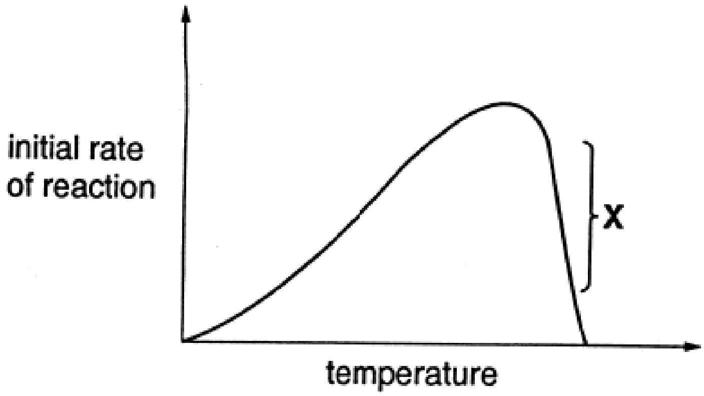
4 A piece of mammalian tissue was homogenised and subjected to differential centrifugation. Four subcellular fractions were investigated to determine their relative biochemical activity. The diagrams indicate the relative activity of certain biochemical processes in these fractions.

Which diagram indicates the fraction with maximum synthesis of messenger RNA?



Nov 97 Q5 Ans B

5 The diagram shows the initial rate of reaction using constant amounts of substrate and enzyme at different temperatures.



What is the reason for the decline in the level of activity in region X?

A breaking of sulfur bridges and ionic bonds in the enzyme
B competition between substrate and product for the active site
C breaking of hydrogen bonds in the enzyme
D insufficient substrate to occupy all the active sites

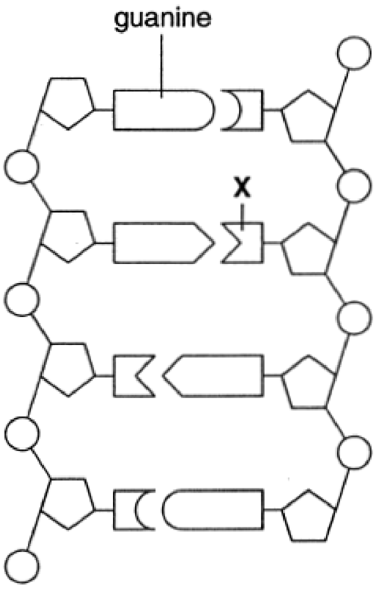
Nov 97 Q9 Ans C

6 Which changes occur when the electron transfer system in animals is inhibited?

	lactate concentration	oxygen consumption	Krebs cycle activity
A	decreased	decreased	increased
B	decreased	increased	increased
C	increased	decreased	decreased
D	increased	increased	decreased

Nov 97 Q28 Ans C

7 The diagram shows part of a molecule of DNA.



What is the name of base X?

A adenine
B cytosine
C thymine
D uracil

Nov 98 Q16 Ans C

8 What is an accurate description of the coding of protein structure by DNA?

A a degenerate, non-overlapping code for the primary structure, determining the location of the folding sites involved in tertiary structure
B a code, with several triplets for each amino acid, determining the shape of the a-helix for the secondary structure
C 64 different complementary codons used in transcription to make the three-dimensional shape of the tertiary structure
D amino acids coded for by three consecutive bases, used with tRNA anticodons in translation to determine the quarternary structure

Nov 04 Q5 Ans A

9 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labeled. These bacteria (generation 1) were then grown in a medium containing only normal nitrogen (^{14}N) for two more generations (generations 2 and 3). The percentage of cells containing ^{15}N in each generation was estimated.

What will be the percentage of cells containing ^{15}N in generations 2 and 3?

percentage of cells containing ^{15}N

	generation 2	**generation 3**
A	100	100
B	100	50
C	50	50
D	50	25

Nov 07 Q7 Ans B

10 The table shows the mRNA codons for 11 different amino acids.

amino acid	mRNA codon
Ala	GCG
Glu	GAG
His	CAC
Leu	CUG CUC

amino acid	mRNA codon
Lys	AAG
Pro	CCU
Thr	ACU
Val	GUG

amino acid	mRNA codon
Arg	CGC
Phe	UUC
Gly	GGA

The first seven DNA triplets coding for a protein are shown below. These are DNA triplets from the coding strand, complementary to the mRNA.

CACGAGGCGAAGGGACACCGC

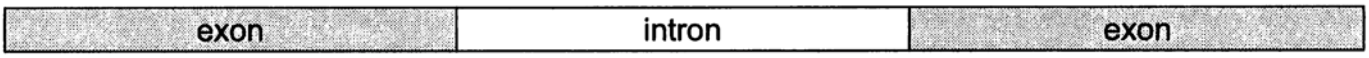
A mutation occurs when the seventeenth nucleotide in the DNA sequence is changed to G.

What is the amino acid sequence for the mutation section of DNA?

A His Glu Ala Lys Gly Arg Arg
B Val Leu Arg Phe Pro Ala Ala
C His Glu Ala Lys Gly His Arg
D Val Leu Arg Phe Pro Val Ala

Nov 07 Q9 Ans B

11 The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

A activation of the gene by a steroid hormone
B attachment of mRNA to a ribosome
C binding of RNA polymerase to DNA
D splicing of pre-mRNA

Nov 07 Q16 Ans D

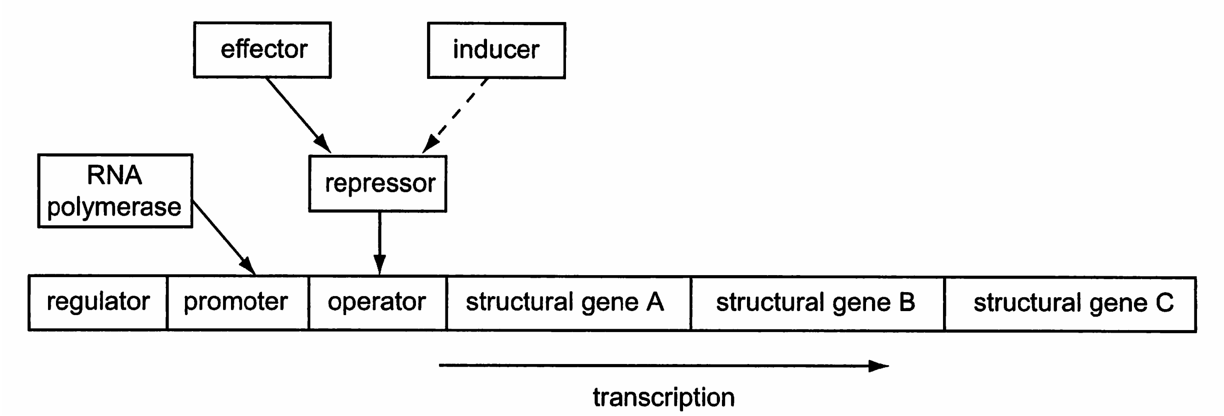
12 One stage during protein synthesis in eukarytoic cells involves preliminary mRNA, known as pre-mRNA. The pre-mRNA is made up of exons and introns.

What describes the next stage in the process?

A All the exons are removed, so that the introns can be translated.
B All the introns are removed, so that the exons can be translated.
C Some of the introns and exons are removed, so that the remaining exons and introns can be translated.
D All the exons and introns are translated.

Nov 07 (H1) Q12 Ans B

13 The diagram shows some of the molecules involved in controlling transcription in bacteria.



RNA polymerase must bind to the promoter to start transcription of the structural genes.
The regulator produces a repressor.
The polypeptide synthesized from the structural gene A acts as an effector.
The effector binds the repressor to the operator to block transcription.
The inducer prevents the repressor binding.

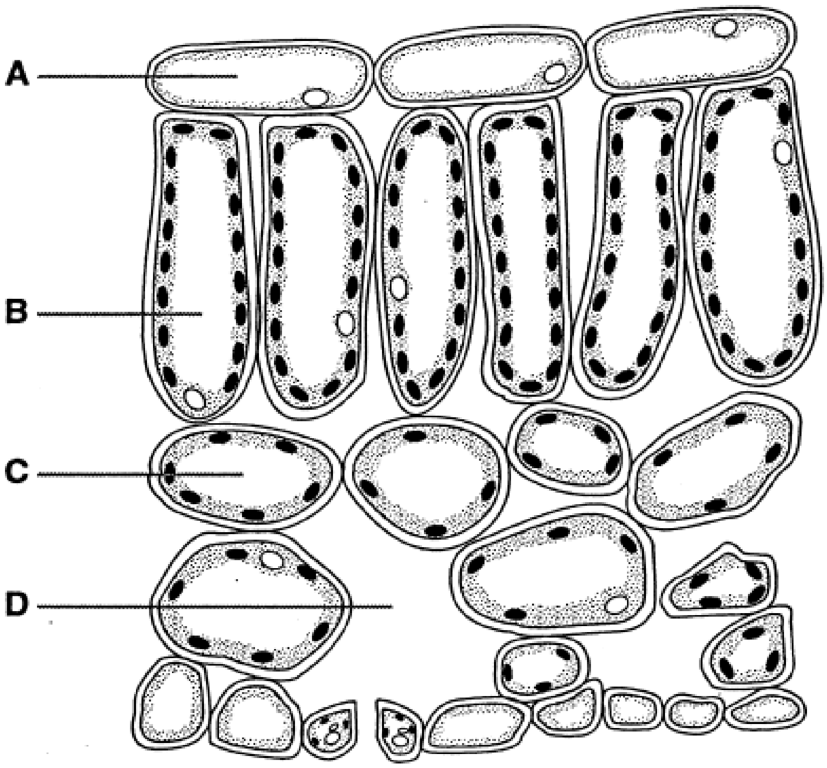
Which molecule acts as an end product inhibitor?

A effector
B inducer
C regulator
D repressor

Nov 08 (H1) Q12 Ans A

14 The diagram represents a transverse section of a leaf.

Which region is the main site of carbon fixation?



Nov 02 Q21 Ans B

15 In additional to ethanol, what does the fermentation of glucose by yeast yield?

	number of molecules		
	CO_2	reduced NAD	ATP
A	0	0	36
B	0	4	2
C	2	0	2
D	2	4	36

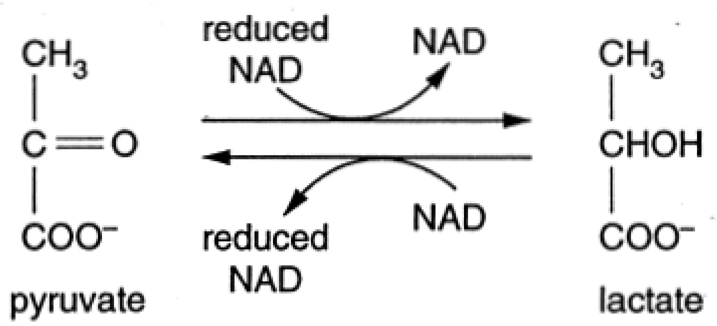
Nov 07 Q27 Ans C

16 What describes oxidative phosphorylation?

A addition of phosphate of ADP using energy gained by transferring electrons along a chain of carriers
B addition of phosphate to ADP using energy gained by transferring electrons between chlorophyll molecules
C addition of phosphate to glucose in the first step of glycolysis
D removal of phosphate from ATP with the release of energy for work within the cell

Nov 07 Q21 Ans A

17 The diagram shows the reversible conversion of pyruvate to lactate by the enzyme lactate dehydrogenase.



What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

A a decrease in cell pH, due to the accumulation of lactic acid
B a decrease in glycolysis, due to the lack of NAD
C an increase in ATP production, due to increased amounts of reduced NAD
D an increase in the activity of the Krebs cycle, due to increased amounts of pyruvate

Nov 98 Q24 Ans B

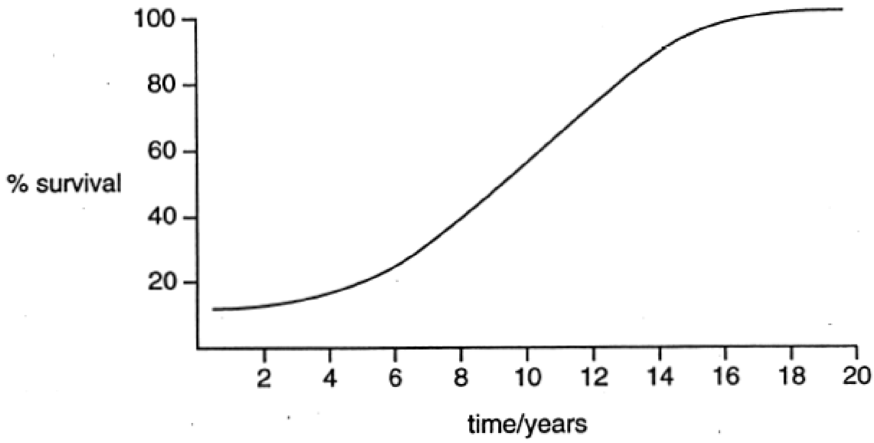
18 At prophase of mitosis, a eukaryote chromosome consists of two chromatids.

What is the structure of a single chromatid?

A one molecule of single-stranded DNA coiled around protein molecules
B two molecules of single-stranded DNA each coiled around protein molecules
C one double helix of DNA coiled around protein molecules
D two double helices of DNA each coiled around protein molecules

Nov 99 Q11 Ans C

19 The graph shows the effect of pesticide treatment on houseflies over a number of years. A standard amount of pesticide was used each year in summer.



How is the effect of the pesticide best explained?

A A few resistant flies reproduced more successfully and the resistance allele increased in frequency.
B At every generation an increasing proportion of flies mutated to become resistant.
C Repeated exposure to the pesticide caused the flies to become more resistant.
D The allele for resistance mutated from the recessive form to the dominant form.

Nov 99 Q20 Ans A

20 Which effect of natural selection is likely to lead to speciation?

A Differences between populations are increased.
B Favourable genotypes are maintained in the population.
C Genetic diversity is reduced.
D Selection pressure on some alleles reduces reproductive success.

Nov 07 Q33 Ans A

21 Potato plants are propagated asexually by tubers. Twenty tubers are collected from one plant and are grown, producing 20 second generation plants. All the tubers from these plants are collected and weighed. The twenty largest are grown under the same conditions as before. All the third generation tubers are collected and are weighed.

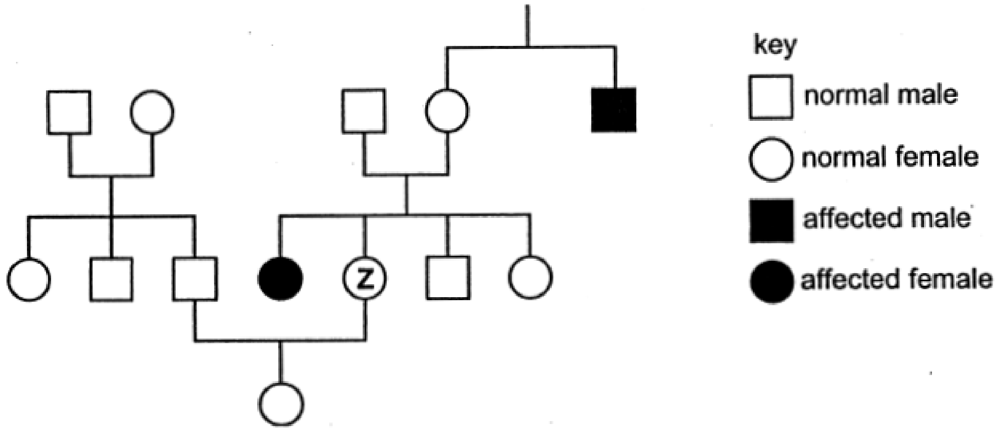
How will the mean mass of these tubers and their genetic variation compare with the second generation?

	third generation tubers	
	mean mass	genetic variation
A	greater	increased
B	greater	unchanged
C	unchanged	reduced
D	unchanged	unchanged

Nov 00 Q 24 Ans D

22 Galactosaemia is a condition caused by a recessive allele of an autosomal gene in which galactose cannot be metabolised.

The diagram shows the pattern of inheritance of galactosaemia in a family.

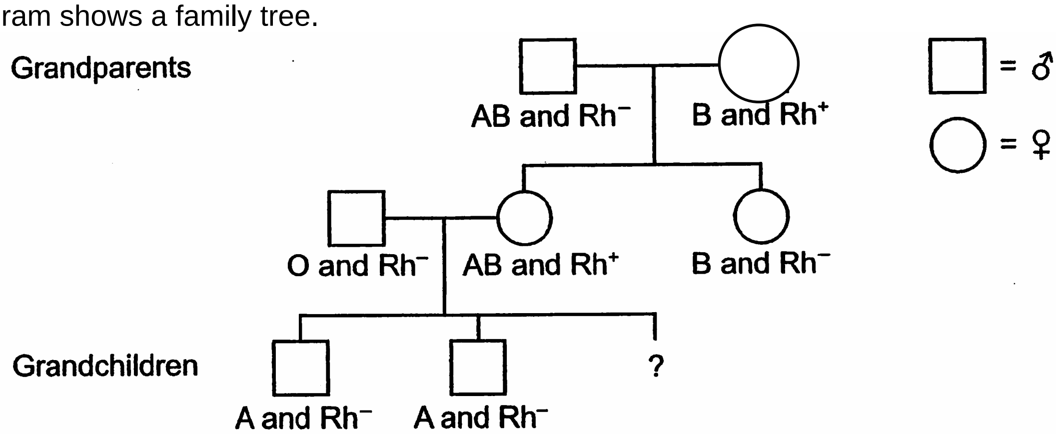


What is the chance that individual Z is a carrier of the condition?

A 1 in 2
B 1 in 3
C 1 in 4
D 2 in 3

Nov 04 Q20 Ans D

23 The diagram shows a family tree.

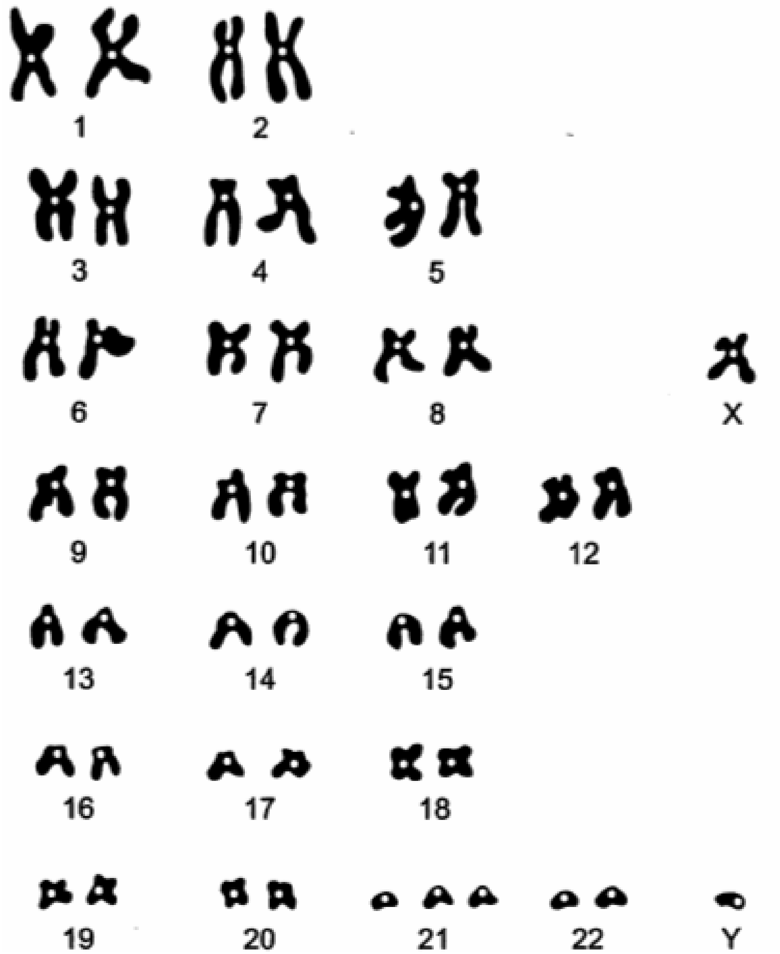


What is the probability that the third grandchild will be a boy with blood group B and Rh positive blood?

A 0.0625 (1 in 16)
B 0.125 (1 in 8)
C 0.25 (1 in 4)
D 0.5 (1 in 2)

Nov 07(H1) Q18 Ans B

24 The diagram shows a human karyotype.



The person who has this karyotype is

A a normal female.
B a female with a chromosome mutation.
C a normal male.
D a male with a chromosome mutation.

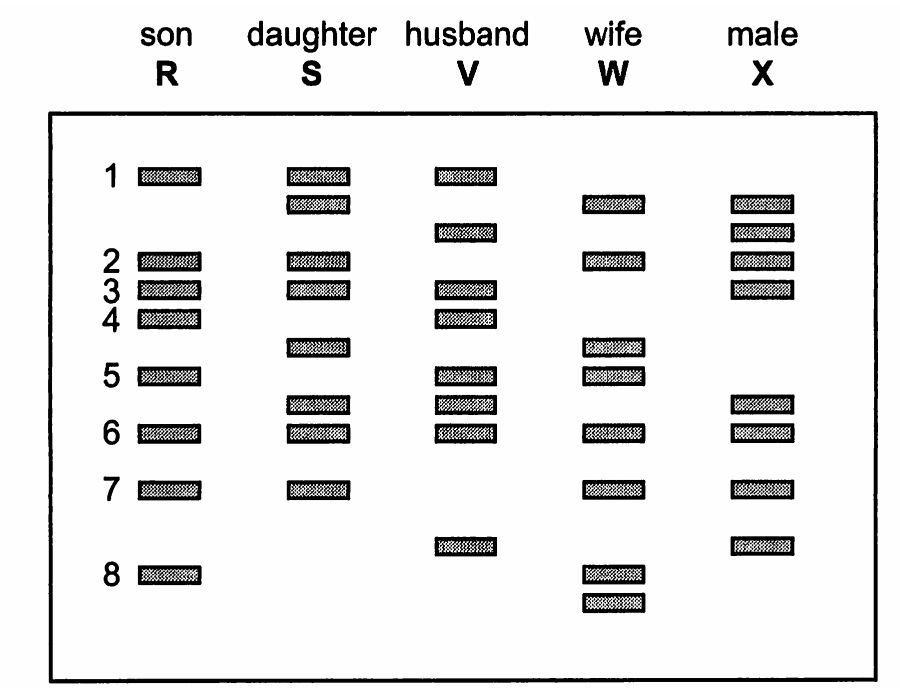
Nov 07(H1) Q16 Ans D

25 If the first three nucleotides in a six-nucleotide restriction site are ATG, what would the next three nucleotides most likely be?

A AGG
B GTC
C CTG
D CAT

Modified from AJC Prelim 08 Q38 Ans D

26 The diagram shows the results of electrophoresis of DNA fragments from four members of a family **R, S, V, W** and another male **X**, who claims that he is actually the father of person **R**.



Which bands on the electrophoretogram indicate that person **V** is the father of person **R**?

A 1 and 2
B 1 and 4
C 2 and 7
D 3 and 6

Nov 08 H1 Q28 Ans B

27 At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to $60^{o}C$.

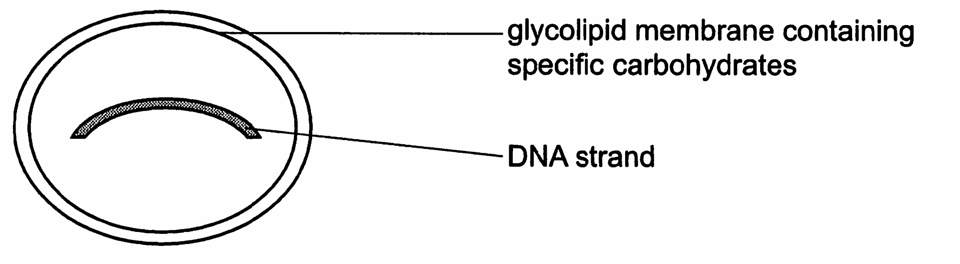
What explains why the denatured DNA strands anneal with primers and **not** each other?

A The primers are shorter and anneal more easily.
B The primers anneal only to the 3' end of the denatured DNA.
C The primer concentration exceeds the denatured DNA concentration.
D The temperature prevents the denatured DNA from annealing together.

Nov 07 Q37 Ans C

28 Gene therapy is a way of treating genetic disease by introducing a piece of DNA into the cells of an affected individual. Liposomes can be used for gene therapy as they target the cells affected by a genetic disease.

The diagram shows the structure of a liposome.



What feature of a cell surface membrane allows the liposome to target cells affected by a genetic disease?

A carrier molecules
B phosphate groups
C protein channels
D receptor molecules

Nov 08 (H1) Q24 Ans D

29 Stem cells are widely used in medical research.

Which property of stem cells makes them particularly useful in this research?

A They can be fused together to form a zygote.
B They can divide and eventually give rise to a whole organism.
C They can divide and be made to differentiate into various types of cell.
D They will continue to divide indefinitely.

Nov 07 (H1) Q29 Ans C

30 Which of these processes could increase crop yield?

P inserting genes for vitamin A production into rice
Q inserting genes for pest resistance into cotton
R inserting genes for herbicide resistance into oilseed rape
S inserting genes for nitrogen fixation into soya beans

A P and Q
B Q and R
C R and S
D S and P

Nov 07 (H1) Q30 Ans B